H-ASIZH: A Highly Reliable Digital Izhikevich Neuron Core Using Adaptive Threshold Control for Fault Accommodation

Ali Roshani, Mohsen Ansari, and Mohammad T. Manzuri Shalmani

***Abstract*—Spiking neural nets scale to microwatts by encoding information in the timing of events but silicon makes them vulnerable to permanent stuck-at faults that trap a neuron in either a firing infinity (digital seizure) or a zero and TMR solves that at $3.40\times$ the cells. This work pushes fault tolerance inside the neuron: H-ASIZH (Homeostatic Affine Scaled Izhikevich) makes the firing threshold a slow state variable whose negative feedback accommodates a permanent fault without an outside observer and a Lyapunov argument guarantees that the mean firing rate is uniformly ultimately bounded against a persistent fault current. 1-cycle Q7.19 core of the placed and routed layout of the SkyWater 130 nm process costs $1.39\times$ the cells at 0.813 pJ per neuron update and brings the dangerous defect set down from 46.2% to 17.5%. Hardening its control plane with a triplicated restoring organ leaves 4.2% of defects dangerous against 12.0% for triplication, at 32% fewer cells, at the same energy per update, and at a $4.76\times$ reliability figure of merit (RFoM) against $1.13\times$.**

***Index Terms*—Spiking neural networks, neuromorphic computing, fault accommodation, fault tolerance, homeostasis, Izhikevich model, Lyapunov stability, gate-level fault injection, energy-efficient VLSI.**

I. Introduction

THE brain-computer interface and neuroprosthetic devices must compute within the body. Doing so requires working under a thermal budget that a standard accelerator cannot meet, and a means of communicating with tissue that already communicates in spikes. Neuromorphic computing accomplishes both simultaneously. Encoding information in the time of events rather than the strength of continuous activations reduces the volume of data that must be moved, enabling low-milliwatt operation while still capturing much of the behaviour of biological circuits [1], [2]; the same behaviour properties also make neuromorphic computing useful for edge-AI workloads that have surpassed von Neumann architectures [3].

Of all the models for digital emulation, a Izhikevich neuron [4] is an unusually good deal: over twenty firing modes from two differential equations for a minuscule fraction of the cost of biophysically detailed models. Multiplierless incarnations are under a thousand transistors [5], [6]; million-neuron fabrics have been fabricated in silicon [7], [8], however none of this settles the question: How does the fabric behave as its transistors die?

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Permanent defects are the concern. Manufacturing defects, electromigration, dielectric breakdown, hot-carrier injection and radiation damage cause a node to short to ground (SA0) or to the supply (SA1) [9]. In a spiking neuron the two polarities fail quite differently. An SA1 in the membrane datapath, or an SA1 in the threshold comparator, acts as a persistent parasitic current: every cycle it drives the membrane to the rail, and it locks the cell into sustained discharge—a *digital seizure* that saturates the spike fabric and flattens an implanted battery in seconds. An SA0, on the other hand, takes out the drive and silences the cell [10]. The difference is important. Saturation almost always kills the network, while silencing is often absorbed in a hidden layer and becomes deadly only at the output [11], [12]. Defect tolerance in VLSI is not new [13], but in its traditional incarnation it assumes a datapath whose correctness is binary—not one that outputs a rhythm.

A. Limits of Conventional and Bio-Inspired Mitigation

The textbook answer is spatial redundancy. Triple Modular Redundancy (TMR) replicates the module to be protected and masks faults with majority voting [14]. Inserting TMR into a neuromorphic substrate it is self-defeating: area and power at least triple, the voter shoves jitter down a channel whose information is in its timing, and tripling the circuit triples the cross-section available to the provoking event [15].

A second line of work has moved towards biological compensation. SPANNER deploys astrocyte-like monitors alongside the cores and modulates synaptic release probability on detection [16]; some distribute astrocytes to restore individual firing rates [17], or make that compensation tunable at run time and report 94% accuracy with 15% of cores faulty [18]. Closest in spirit to this paper, Johnson *et al.* use homeostatic regulation to provide fault tolerance, but the underlying mechanism derives from a distributed astrocyte net rather than directly from the neuron's own state [19]. Others restore execution within an already-faulty accelerator [20], or seek structural containment within a single dendritic branch [21], or within a single spiking pixel [22]. Two more approaches complete the list: temporal redundancy swaps replication for re-execution [23], and mapping schemes move the computation itself either away from an unreliable memory [24], or between diverse heterogeneous cores [25].

B. The Research Gap

Each of these advances is real, and they all lead to the same place. Six remaining gaps together constitute the surface on

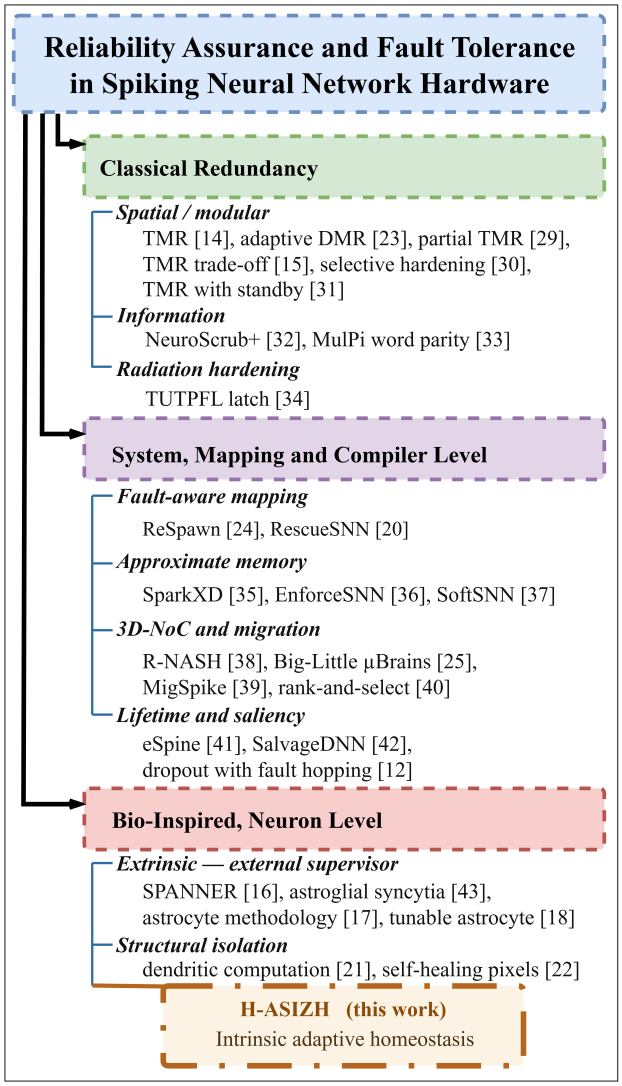

Fig. 1. Taxonomy of SNN fault-tolerance work by the level at which the mitigation acts. H-ASIZH occupies the remaining leaf, intrinsic to the neuron's own dynamics.

which this paper stands.

The redundancy trap. Spatial redundancy pays in area and power [14], [15]; temporal redundancy pays in latency and energy [23], and against a permanent defect it achieves nothing, since a fault on every cycle provides the same wrong result every time.

Loss of determinism. Fault-aware mapping and migration make a spike's path specific to one specific defect map of one specific die [24], [25]. Arrivals can be re-analysed afterwards but not guaranteed in advance, and for a prosthesis that looped in less than a millisecond, a bound that holds typically is not good enough.

The unwatched watcher. Bio-inspired schemes avoid redundancy by adding a supervisor [16]–[18] – one made from the same gates, on the same die, at the same defect density as the circuit that it is intended to rescue. A permanent fault in its configuration state doesn't just incapacitate the protection – it inverts it.

The price of endogenous resilience. The one family that does inject resilience within the neuron does so structurally, copying each dendritic compartment's arithmetic with overhead of around 60% per neuron [21]. And the deeper issue is direction: the field's centre of gravity still resides in a top-down manner – treating the neuron core as a primitive to route around rather than something to harden.

Heuristic rather than formal guarantees. Robustness is typically estimated based on the campaign, and the lack of a counterexample does not say what might happen under untested conditions. What is missing is a stability argument that bounds the firing rate for every admissible parasitic current [26], [27].

Assessment that is not physically faithful. Assertions are largely made based on high-level simulations, or gate-level campaigns that are limited to two-valued logic, which resolves an unknown optimistically in a way that a real comparator or a select line would not. Existing frameworks either operate at a level above the netlist or constrain the design to a single FPGA target, so the coverage reported describes the model or the device rather than the circuit [28].

Fig. 1 categorizes this space by the layer of mitigation, and displays a space that has been explored virtually completely at every layer but one. Classical redundancy was refined well past triplication alone [29]–[31], and has been extended by information redundancy, from approximate scrubbing [32] to upset-tolerant cells [33] and in-memory substrates [34]. At the system level, the focus is on error absorption; approximate memories give up a tunable error budget for energy savings [35]–[37], 3-D networks on chip reroute around damaged tiles [38], and migration, remapping, and lifetime-aware placement push computation onto healthy cores [39]–[42]. The bio-inspired work has also seen network scale work [43]. Each one of these acts on, around, or above the neuron. None of them acts inside the neuron.

C. Proposed Solution and Contributions

We close all six gaps with a solitary architectural commitment: the fault-tolerance is implemented as part of the neuron's own state equations. The neuron we propose, H-ASIZH (Homeostatic Affine Scaled Izhikevich), is an intrinsic controller, which treats the firing threshold as a slow state variable that can be elevated to absorb the energy of a permanent fault. No external guardian is left to break, because the regulator and control plant are the same circuit, and because nothing migrates, the fabric remains as deterministic when faulty as it was when not.

- **Reformulation and stability proof.** An affine map carries the Izhikevich equations into a compact, shift-only domain with no approximation, reproducing nine firing patterns to within 10^{-13} RMSE. With the controller embedded in the mapped dynamics, we prove by Lyapunov analysis that the mean firing rate is Uniformly Ultimately Bounded under any persistent fault current, the residual band being set by the spike penalty rather than by the defect amplitude. Boundedness of the rate and boundedness of the machine word are distinct guarantees: the theorem supplies the first, the clamping network of Section IV-C the second.

- **Hardware-aware coefficient optimisation.** Multi-objective design space exploration binds the controller parameters to an exact power of two, which simplifies the resilience arithmetic to a shift-and-add.
- **RTL micro-architecture.** A 1-cycle unrolled Q7.19 datapath with the 4-segment piecewise-linear estimator (implemented without multiplier), a clamping network, a fast-bleed recovery path, and a proportional β-brake - mitigation performed entirely in the neuron core, leaving routing and schedule untouched.
- **Physically faithful fault analysis.** A technology-agnostic gate-level injection framework applied to the synthesised netlist using four-valued logic that compiles the fault into the circuit, rather than forcing it out of the simulator as open frameworks do [44], with an exhaustive campaign of five architectures. The resulting projection reveals the proposed core outperforming full triplication by more than a factor of 4 at one-third less silicon.[1]

Section II develops the mapping and the controller; Section III-A establishes stability; Section IV describes the micro-architecture; Section V the injection framework; Section VI presents an on-chip learning case study; and Section VII reports the results.

II. Mathematical Framework and Hardware-Aware Affine Mapping

A. Baseline Model and Its VLSI Obstacles

The Izhikevich model [4] reduces neuronal dynamics to two equations in the membrane potential v and a slow recovery variable u:

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I \tag{1}$$

$$\frac{du}{dt} = a(bv - u) \tag{2}$$

with the discontinuous after-spike reset

$$\text{if } v \geq 30 \text{ mV}: \quad v \leftarrow c, \quad u \leftarrow u + d \tag{3}$$

where I is the synaptic current and a, b, c, d select the firing pattern.

Two properties obstruct direct digital implementation: the potential is signed, resting near -65 mV and peaking at $+30$ mV; and the quadratic term demands multipliers, which dominate area and power in a many-core array. In-memory substrates evade the multiplier by computing inside the storage macro [34], but surrender the bit-exact reproducibility a gate-level campaign depends on.

B. Affine Transformation and Scaling

We map the state variables into a compact, predominantly positive fixed-point domain, shifting the potential by $\eta = 65$ to place rest at the hardware origin and contracting by $\sigma = 2^{-6}$:

$$V_{hw} = \frac{v_{bio} + 65}{64} \tag{4}$$

with $U_{hw} = u_{bio}/64$ and $I_{hw} = I_{bio}/64$ taking the same scaling without offset. The contraction is the motive, not a side effect: division by 2^6 is a right-shift costing no logic, and it confines the trajectory to roughly $[-0.5, 1.5]$, leaving the integer field almost empty so the released bits become guard bits that absorb an overshoot or an injected defect without wrapping the accumulator.

[1]Framework, fault dictionaries and campaign scripts: https://github.com/aliroshanidino/GLFI_fault_campaign.

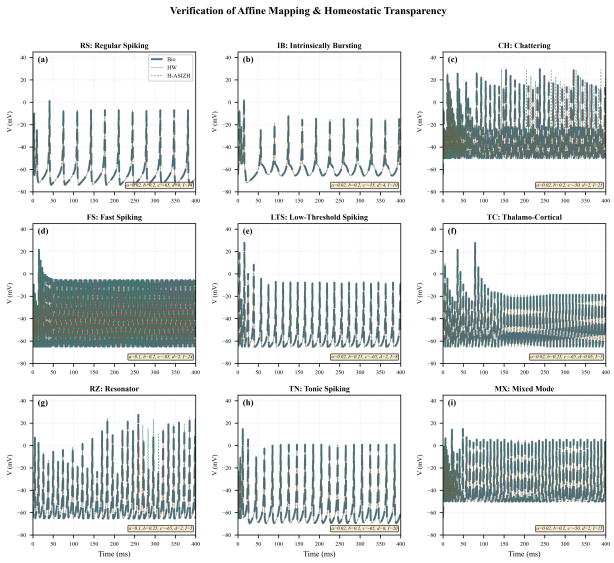

Fig. 2. Spike-train overlay of the floating-point biological model and the affine-mapped hardware model across nine firing patterns.

C. Algebraic Equivalence

Substituting $v_{bio} = 64V_{hw} - 65$ into (1) and dividing through by 64 gives the hardware-domain dynamics directly:

$$\frac{dV_{hw}}{dt} = 2.56\, V_{hw}^2 - 0.2\, V_{hw} + I_{hw} - U_{hw} + \Psi_{bias} \tag{5}$$

with $\Psi_{bias} = -0.25$ collecting the constant terms, and

$$\frac{dU_{hw}}{dt} = a\big(bV_{hw} - U_{hw} - \tfrac{65}{64}b\big) \tag{6}$$

The transformation is exact, so nullclines and phase-portrait topology are preserved—unlike comb-exponential [6] and hyperbolic [5] substitutions, which buy multiplier-free operation at the cost of residual model error. Parameters map as $c_{hw} = (c + 65)/64$, $d_{hw} = d/64$, with a, b invariant (Table I); Fig. 2 overlays the two models at RMSE below 10^{-13}.

D. The Intrinsic Homeostatic Controller

Prior designs rely on extrinsic supervisors—modelled astrocytes monitoring activity and modulating weights [16]—or relocate computation away from a damaged core [39], presuming healthy spare capacity. Our route is control-theoretic. Fault-tolerant control distinguishes *masking*, which conceals a defect by replication, from *accommodation*, in which the regulator concedes the defect is permanent and retunes a slow set-point under output feedback until the loop settles on a new admissible equilibrium. The only quantity inside a neuron that directly gates the regulated output is the firing threshold, so we make it a slow state variable $\theta(t)$ resting at $\theta_0 = 1.1$ ($\equiv +5.4$ mV)—on the regenerative upstroke, just below the divergent PWL segment, while the apex $+30$ mV ($\equiv 1.484375$) remains the saturation clamp.

TABLE I
BASELINE AND HARDWARE-MAPPED IZHIKEVICH PARAMETERS (a, b INVARIANT UNDER THE MAP; MAPPED VALUES TO THREE DECIMALS). RS REGULAR SPIKING, IB INTRINSICALLY BURSTING, CH CHATTERING, FS FAST SPIKING, LTS LOW-THRESHOLD SPIKING, TC THALAMO-CORTICAL, RZ RESONATOR, TN TONIC, MX MIXED MODE

	Biological domain					Hardware domain		
Pattern	a	b	c	d	I_{app}	c_{hw}	d_{hw}	I_{hw}
RS	0.02	0.20	−65	8.00	14	0	0.125	0.219
IB	0.02	0.20	−55	4.00	10	0.156	0.063	0.156
CH	0.02	0.20	−50	2.00	25	0.234	0.031	0.391
FS	0.10	0.20	−65	2.00	24	0	0.031	0.375
LTS	0.02	0.25	−65	2.00	8	0	0.031	0.125
TC	0.02	0.25	−65	0.05	5	0	0.001	0.078
RZ	0.10	0.25	−65	2.00	5	0	0.031	0.078
TN	0.02	0.20	−65	6.00	20	0	0.094	0.313
MX	0.02	0.20	−50	2.00	15	0.234	0.031	0.234

A permanent stuck-at-1 defect acts as a persistent parasitic current driving the neuron into a digital seizure. Three parameters absorb it: the leak α returns θ to θ_0 absent rapid firing, keeping the defence transparent when healthy; the penalty γ raises θ at every spike, so fault-induced discharge chokes itself; and the brake β amplifies the reset jump of U_{hw} in proportion to $\theta - \theta_0$. The complete system is

$$\frac{dV_{hw}}{dt} = 2.56\,V_{hw}^2 - 0.2\,V_{hw} + I_{hw} - U_{hw} + \Psi_{bias} \tag{7}$$

$$\frac{dU_{hw}}{dt} = a\big(bV_{hw} - U_{hw} - \tfrac{65}{64}b\big) \tag{8}$$

$$\frac{d\theta}{dt} = \alpha(\theta_0 - \theta) + \gamma \sum_k \delta(t - t_k) \tag{9}$$

with the reset condition

$$\text{if } V_{hw} \geq \theta: \quad \begin{cases} V_{hw} \leftarrow c_{hw} \\ U_{hw} \leftarrow U_{hw} + d_{hw} + \beta(\theta - \theta_0) \end{cases} \tag{10}$$

The three parameters map term by term onto intracellular calcium handling, so the silicon that buys fault tolerance also materialises the state local on-chip learning requires [19].

E. Hardware-Aware Design Space Exploration

Lyapunov analysis bounds the firing rate for any positive α, γ, but says nothing about recovery speed or silicon cost. Keeping the datapath multiplier-free confines all three coefficients to powers of two, collapsing the search onto a grid: $\alpha \in [2^{-10}, 2^{-3}]$, $\gamma \in [2^{-6}, 2^{0}]$, $\beta \in [2^{0}, 2^{6}]$. Cost thus enters as a structural constraint—one shifter per parameter—leaving the objective to score behaviour alone.

That objective must be neuro-metric: a controller shifting every spike by two cycles has not failed, whereas one preserving the count while destroying the rhythm has. Per pattern p we score four trains—golden G_p and unprotected-faulted U_p, computed once, plus healthy-protected H_p and faulted-protected F_p per candidate—over three disjoint windows: $\mathcal{W}_1 = [0, T]$, $\mathcal{W}_2 = [T_0, T_1]$ during the disturbance, and $\mathcal{W}_3 = [T_1, T]$ after it. The third is indispensable, because a controller can suppress a seizure perfectly and still leave the neuron inert once the defect clears. With $N_{\mathcal{W}}$ the spike count and $\overline{\text{ISI}}_{\mathcal{W}}$ the mean interval,

$$\begin{aligned} \Delta_{\mathcal{W}}(S, R) = \frac{1}{2}\left|\frac{N_{\mathcal{W}}(S) - N_{\mathcal{W}}(R)}{N_{\mathcal{W}}(R)}\right| \\ + \frac{1}{2}\left|\frac{\overline{\text{ISI}}_{\mathcal{W}}(S) - \overline{\text{ISI}}_{\mathcal{W}}(R)}{\overline{\text{ISI}}_{\mathcal{W}}(R)}\right| \end{aligned} \tag{11}$$

with $\Delta_{\mathcal{W}} = 1$ when S has fewer than two spikes—the same value a controller halving or doubling the rate incurs, so silence is charged as a gross rate error rather than treated as free. Two terms follow,

$$c_{\text{bio}} = \Delta_{\mathcal{W}_1}(H_p, G_p), \qquad c_{\text{rec}} = \Delta_{\mathcal{W}_3}(F_p, G_p) \tag{12}$$

enforcing transparency when healthy and recovery after the defect, while the suppression term takes the size of the seizure itself as denominator:

$$c_{\text{sup}} = \left|\frac{N_{\mathcal{W}_2}(F_p) - N_{\mathcal{W}_2}(G_p)}{N_{\mathcal{W}_2}(U_p) - N_{\mathcal{W}_2}(G_p)}\right| \tag{13}$$

which makes it interpretable: 0 is complete suppression, 1 a controller achieving nothing, and $c_{\text{sup}} > 1$ one actively worse than no protection—a region that is not hypothetical. Since ϵ is chosen large enough that an unprotected cell seizes for every pattern, the denominator is bounded away from zero by construction. The objective aggregates over the set,

$$(\alpha^\star, \gamma^\star, \beta^\star) = \underset{\mathcal{S}_\alpha \times \mathcal{S}_\gamma \times \mathcal{S}_\beta}{\arg\min} \sum_{p \in \mathcal{P}} \big[w_1 c_{\text{bio}} + w_2 c_{\text{sup}} + w_3 c_{\text{rec}}\big] \tag{14}$$

with $w_1 = 0.25$, $w_2 = 0.40$, $w_3 = 0.35$. Summing over $\mathcal{P}$ is a hardware requirement: one set of shift constants is hardwired and must serve every pattern, so a triplet excellent for one and destructive for another is inadmissible. Algorithm 1 enumerates the grid over $T = 6000$ cycles with $[T_0, T_1] = [700, 2700]$ and disturbance $\epsilon = 85/64$, integrating at a fixed Euler step of 2^{-6}. No term refers to an expected solution, so the reported triplet is a property of the objective rather than of the initialisation.

The exploration returns $\alpha^\star = 2^{-6}$, $\gamma^\star = 2^{-3}$, $\beta^\star = 2^{3}$, realised as $\gg 6$, $\gg 3$ and $\ll 3$: three shifters, two adders and

Algorithm 1 Hardware-Aware Design Space Exploration for the H-ASIZH Controller

Require: power-of-two grids $\mathcal{S}_\alpha, \mathcal{S}_\gamma, \mathcal{S}_\beta$; pattern set $\mathcal{P}$; horizon T; fault window $[T_0, T_1]$; disturbance ϵ; weights w_1, w_2, w_3
Ensure: optimal triplet $(\alpha^\star, \gamma^\star, \beta^\star)$

for all $p \in \mathcal{P}$ **do** ▷ references, computed once
  $G_p \leftarrow \text{RUN}(p,\ \text{controller off},\ 0)$
  $U_p \leftarrow \text{RUN}(p,\ \text{controller off},\ \epsilon)$
end for
$C^\star \leftarrow +\infty$
for all $(\alpha, \gamma, \beta) \in \mathcal{S}_\alpha \times \mathcal{S}_\gamma \times \mathcal{S}_\beta$ **do**
  $C \leftarrow 0$
  for all $p \in \mathcal{P}$ **do**
    $H_p \leftarrow \text{RUN}(p,\ (\alpha, \gamma, \beta),\ 0)$
    $F_p \leftarrow \text{RUN}(p,\ (\alpha, \gamma, \beta),\ \epsilon)$
    $c_{\text{bio}} \leftarrow \Delta_{[0,T]}(H_p, G_p)$ ▷ transparency
    $c_{\text{sup}} \leftarrow$ evaluate (13) on $[T_0, T_1]$
    $c_{\text{rec}} \leftarrow \Delta_{[T_1,T]}(F_p, G_p)$ ▷ recovery
    $C \leftarrow C + w_1 c_{\text{bio}} + w_2 c_{\text{sup}} + w_3 c_{\text{rec}}$
  end for
  if $C < C^\star$ **then**
    $C^\star \leftarrow C;\ (\alpha^\star, \gamma^\star, \beta^\star) \leftarrow (\alpha, \gamma, \beta)$
  end if
end for
return $(\alpha^\star, \gamma^\star, \beta^\star)$

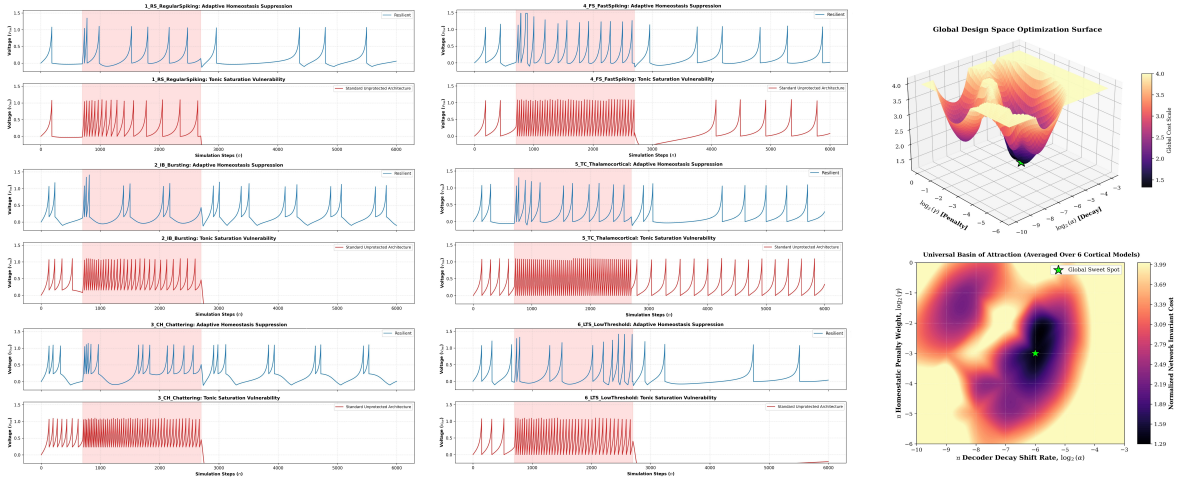

Fig. 3. Cost landscape over the (α, γ) subspace at $\beta = \beta^\star$. The two high-cost ridges are self-silencing of a healthy neuron and failure to contain the seizure, separated by the valley holding the selected operating point.

one comparator, with no multiplier on the critical path. Fig. 3 shows why the valley is broad in α and narrow in γ—small α with large γ ratchets the threshold up faster than it relaxes and the neuron suffocates, while large α with small γ lets it decay before the penalty accumulates.

Two results emerge. The brake has a genuine optimum: suppression cost falls from $\beta = 2^0$ to 2^3, flattens at 2^4, then deteriorates past unity at 2^6, where the protected neuron discharges more heavily than an unprotected one. The mechanism is resonance rather than overshoot—an excessive brake drives the membrane far below rest at every reset and trips the emergency discharge path of Section IV. And threshold adaptation alone is insufficient: with that path disabled, only one pattern in six resumes firing after the disturbance, at every β, because each seizure spike adds $\beta(\theta - \theta_0)$ to the recovery variable and the accumulation relaxes only with $a \approx 0.02$. The neuron survives the seizure and is extinguished by its own defence. Bounded firing and functional survival are not the same property.

III. Validation: Stability Analysis and Phase-Space Dynamics

A. Lyapunov Stability Analysis and Uniform Ultimate Boundedness (UUB)

The objective is to show, within nonlinear control theory [26], [27], that under a permanent hardware failure modelled as a persistent parasitic current I_{fault} the internal feedback loop prevents exponential state divergence, so the firing frequency remains Uniformly Ultimately Bounded (UUB).

Because the controller includes a discontinuous term driven by Dirac pulses ($\sum_k \delta(t - t_k)$), classical continuous Lyapunov stability does not apply directly. We therefore invoke the Averaging Theorem for hybrid systems, mapping the discrete spike train onto a continuous mean firing rate $r(t)$. Averaging is legitimate because θ is the slowest state: at $\alpha = 2^{-6}$ per clock cycle it moves by a negligible fraction of its range between two consecutive spikes, so the membrane pair (V_{hw}, U_{hw}) may be treated as having reached its periodic orbit for the instantaneous θ:

$$\frac{d\theta}{dt} = \alpha(\theta_0 - \theta) + \gamma r(t) \tag{15}$$

The prerequisite is a monotonic relation between threshold and firing frequency. The inter-spike interval follows from integrating the membrane equation from the reset voltage c_{hw} to the instantaneous threshold θ, with $\dot{V}_{hw}$ the right-hand side of (5) and the recovery variable held at its slowly varying value:

$$T_{ISI} = \int_{c_{hw}}^{\theta} \frac{dV_{hw}}{\dot{V}_{hw} + I_{fault}} \tag{16}$$

Since $r = 1/T_{ISI}$, raising the upper limit θ monotonically increases T_{ISI} and strictly reduces the firing frequency:

$$\frac{\partial r}{\partial \theta} < 0 \tag{17}$$

Let θ^* be the unique homeostatic equilibrium threshold under a permanent defect I_{fault}:

$$\alpha(\theta_0 - \theta^*) + \gamma r^* = 0 \tag{18}$$

The candidate $\mathcal{V}(\theta) = \frac{1}{2}(\theta - \theta^*)^2$ is strictly positive-definite, vanishing only at θ^*. Its Lie derivative along the state trajectories is

$$\begin{aligned} \dot{\mathcal{V}}(\theta) &= \frac{\partial \mathcal{V}}{\partial \theta}\frac{d\theta}{dt} \\ &= (\theta - \theta^*)\left[\alpha(\theta_0 - \theta) + \gamma r(t)\right] \end{aligned} \tag{19}$$

Substituting the equilibrium condition $\alpha\theta_0 = \alpha\theta^* - \gamma r^*$:

$$\dot{\mathcal{V}}(\theta) = (\theta - \theta^*)\big[-\alpha(\theta - \theta^*) + \gamma\big(r(t) - r^*\big)\big] \tag{20}$$

Two regimes emerge. In the *hyperactive* phase ($\theta < \theta^*$) an abnormal outburst gives $r(t) > r^*$ by (17); with $(\theta - \theta^*)$ negative and $(r - r^*)$ positive, $\dot{\mathcal{V}} < 0$. In the *hypoactive* phase ($\theta > \theta^*$) the threshold over-corrects, $r(t) < r^*$, and the positive $(\theta - \theta^*)$ multiplied by a negative bracket again yields $\dot{\mathcal{V}} < 0$. Since $\dot{\mathcal{V}}$ is strictly negative-definite across all non-equilibrium states, θ converges asymptotically to θ^* and $r(t)$ stays bounded, preventing digital saturation. ■

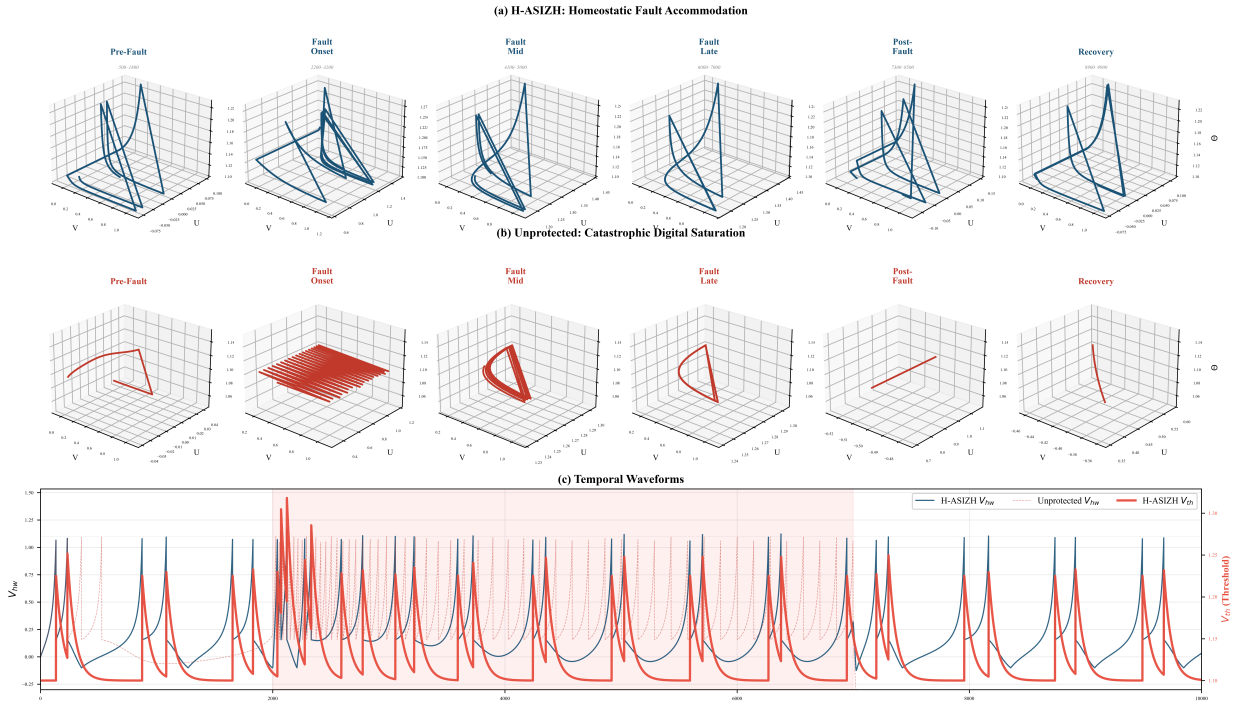

Fig. 4. Phase-space dynamics: (a) transition from the nominal 2D limit cycle to a bounded homeostatic attractor upon fault injection; (b) membrane trace showing graceful degradation; (c) autonomous threshold elevation absorbing the defect.

This establishes convergence of the *averaged* dynamics. The physical system is hybrid—θ takes a discrete increment γ at every firing instant and relaxes between them—so it cannot settle onto θ^* exactly but enters and remains within a residual band $|\theta - \theta^*| \lesssim \gamma$. The hardware guarantee is therefore uniform ultimate boundedness with a bound proportional to γ and independent of the disturbance amplitude: a larger defect displaces θ^* but does not widen the oscillation around it, so the firing rate degrades predictably rather than catastrophically.

B. Phase-Space Interpretation

The mechanism admits a geometric reading in (V_{hw}, U_{hw}, θ). Fault-free, θ stays at θ_0 and the state is confined to a planar slice, executing a canonical limit cycle. A stuck-at-1 defect on a high-order accumulator bit warps that vector field; in an unprotected core it destroys the limit cycle and the trajectory accelerates into saturation. H-ASIZH prevents the collapse by making θ an active third dimension: the defect-induced burst activates the penalty term, θ expands along that axis and lifts the trajectory out of the compromised plane, deforming the integration boundary and absorbing the injected energy, while the β-brake deepens the reset along U_{hw} and adds damping. Instead of diverging, the trajectory settles into a bounded three-dimensional limit cycle—an adaptive homeostatic attractor (Fig. 4).

IV. Hardware-Aware Micro-Architecture

The H-ASIZH core is described in VHDL as a flat, 1-cycle unrolled pipeline, which eliminates the latency jitter of multi-cycle execution: in a fabric that encodes information in spike timing, an update latency varying with operand values corrupts the encoding. The datapath (Fig. 6) splits into a combinational EX stage carrying the arithmetic and a WB stage enforcing the structural bounds, with three 26-bit state registers—V-REG, U-REG and THETA-REG—at their boundary.

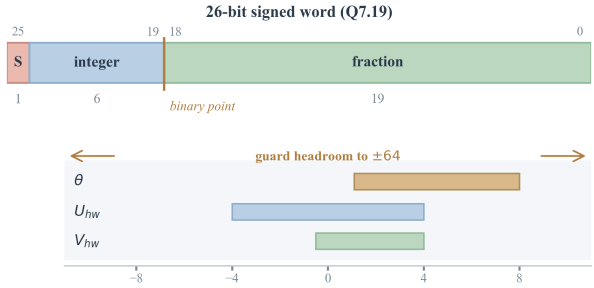

Fig. 5. The Q7.19 datapath word. Top: bit allocation, binary point after bit 19. Bottom: operating envelope of the three state variables against the representable range, showing the guard headroom before wrap-around.

A. The Fixed-Point Word and Its Guard Bits

The datapath carries a 26-bit signed word in Q7.19 format—one sign bit, six integer bits, nineteen fractional bits (Fig. 5). Two consequences of the affine map make that word sufficient where floating point would otherwise be required.

The first is headroom. With $\sigma = 2^{-6}$, V_{hw} occupies $[-0.5, 1.5]$ nominally and is clamped at 4.0, U_{hw} is bounded by ± 4.0 and θ spans 1.1 to 8.0, whereas Q7.19 represents ± 64: every state variable lives within roughly one eighth of the available magnitude. This matters because two's-complement overflow inverts rather than saturates, turning a membrane on the verge of a spike into one in deep hyperpolarisation in a single clock edge and voiding the argument of Section III-A.

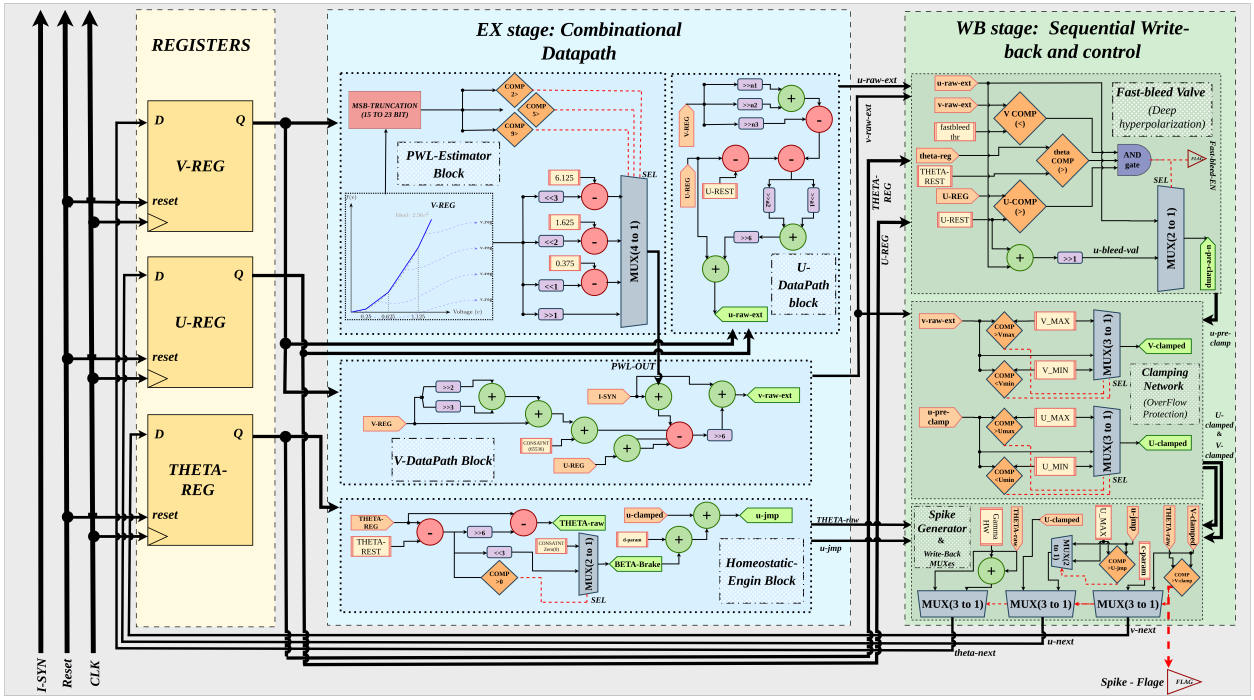

Fig. 6. H-ASIZH micro-architecture. The EX stage holds the PWL estimator, the V- and U-datapaths and the homeostatic engine; the WB stage holds the clamping network, the fast-bleed valve and the spike generator. All paths resolve within one clock period.

The second is resolution. The smallest quantity that must accumulate without vanishing is a threshold decrement of order $2^{-6}(\theta-\theta_0)$; nineteen fractional bits put the quantisation floor at $2^{-19} \approx 1.9 \times 10^{-6}$, three orders of magnitude below it. Widening the word would improve neither margin, since the dominant error is the piecewise-linear estimate rather than quantisation.

B. Multiplier-Free PWL Estimator

The quadratic term $2.56\,V_{hw}^2$ is the one operation that would otherwise demand a 26-bit multiplier per neuron. It is approximated by a four-segment piecewise-linear estimator whose slopes are exact powers of two, so each segment is a wired shift and a constant subtraction:

$$\mathrm{PWL}(V) = \begin{cases} 0, & V < 0 \\ V \gg 1, & 0 \le V < 0.25 \\ (V \ll 1) - 0.375, & 0.25 \le V < 0.625 \\ (V \ll 2) - 1.625, & 0.625 \le V < 1.125 \\ (V \ll 3) - 6.125, & V \ge 1.125 \end{cases} \tag{21}$$

The leading branch is a design decision rather than a limitation. The clamping network of Section IV-C holds V_{hw} at or above -0.5, so the sub-rest excursion is bounded and the true quadratic contributes at most 0.64 there, against a derivative dominated by the leak and the recovery variable. Forcing the estimator to zero below rest costs one sign-bit comparison, keeps the regenerative branch monotone, and removes any possibility that an arithmetic right shift returns a negative value for a term positive by construction. The selector reads the sign bit with bits 15–23 of V-REG; three breakpoint comparators and that sign test drive a 5-to-1 multiplexer, so the nonlinear evaluation costs one multiplexer delay and no arithmetic.

C. Datapaths, Homeostatic Engine and Write-Back

Two adder trees run concurrently, and every coefficient in them is a sum of powers of two, so the whole recurrence is shifts and additions. The V-datapath forms

$$\begin{aligned} \Delta V &= \mathrm{PWL}(V) + I_{hw} - \big[(V \gg 3) + (V \gg 6)\big] - U - 0.25, \\ V^+ &= V + (\Delta V \gg 6), \end{aligned} \tag{22}$$

where $(V \gg 3) + (V \gg 6) = \frac{9}{64}V$ realises the linear term of (5) in two shifts and $\gg 6$ is the Euler step $\Delta t = 2^{-6}$. The U-datapath forms

$$\begin{aligned} \Delta U &= (V \gg 2) - (V \gg 4) + (V \gg 6) - U_{\mathrm{rest}} - U, \\ U^+ &= U + \big[(\Delta U \gg 6) + (\Delta U \gg 8)\big] \gg 6, \end{aligned} \tag{23}$$

in which the three shifts on V realise $b = \frac{13}{64}$, the two on ΔU realise $a = \frac{5}{256}$, and $U_{\mathrm{rest}} = -\frac{13}{64}$ is the recovery variable at rest. The deviations from the ideal coefficients are those a shift-only datapath admits by construction, and the controller was tuned on these realised dynamics. A third branch implements the controller: a subtractor forms $\theta - \theta_0$, a $\gg 6$ realises the α leak, and the same error is rectified and shifted $\ll 3$ to form the β brake.

Because one set of shift constants is hardwired, the core realises the $b = 0.20$ family of Table I—RS, IB, CH, FS, TN and MX. The nine patterns of Fig. 2 therefore validate the *affine mapping*; the six validate the *implemented core*, and every hardware result here is taken over that set.

The WB stage enforces in hardware the boundedness proven analytically: parallel comparators clamp `v-raw-ext` and

`u-raw-ext` to their saturation limits, so the accumulator can never wrap. A *fast-bleed valve* addresses the opposite risk—once a seizure is suppressed, the inhibitory charge in U-REG can trap the neuron in deep hyperpolarisation—by bypassing the nominal integration and driving the recovery variable halfway back to rest in one cycle. Without it, five of six patterns never resume after the defect clears. A 26-bit comparator then evaluates the clamped voltage against the decayed threshold, and the spike flag steers the write-back multiplexers, routing V to c_{hw}, adding d_{hw} and the β brake to U, and penalising θ by γ.

D. Control-Plane Hardening: Light TMR and the Restoring Organ

Accommodation defends the wide arithmetic well, because a defect there perturbs a quantity the loop measures and can absorb. It cannot defend the loop's own decision logic: a stuck-at on the spike comparator, a clamp flag or THETA-REG corrupts the signal the feedback is computed from, and no threshold elevation recovers a decision never taken. Two variants therefore harden the control plane and leave the datapath to the controller.

Light TMR triplicates the decision logic only. THETA-REG becomes three 26-bit registers reduced by a bitwise majority voter, $\theta_v = (\theta_A \wedge \theta_B) \vee (\theta_B \wedge \theta_C) \vee (\theta_A \wedge \theta_C)$, and seven single-bit decisions—the spike flag, brake-enable, fast-bleed enable and four saturation flags—are triplicated the same way. Each replica derives its flag from its own threshold copy, so a defect in one register is outvoted rather than propagated, while V-REG, U-REG, the PWL estimator and both adder trees remain simplex. The cost is $1\,500$ cells and a 3.6% longer critical path.

The restoring organ closes the gap Light TMR leaves: a single voter reduces three replicas to one net, and that net is a single point of failure downstream of everything it protects. Following the classical construction, this variant triplicates the voters themselves, so no reduction to a single net occurs before the register boundary. The 124 cells it adds—about 3%—remove the last unreplicated decision node, and Section VII-B shows what they buy.

V. A Generic Gate-Level Fault Injection Framework

The stability argument of Section III-A is valid for a continuous disturbance on an idealised state; silicon does not break that way. A permanent defect damages a single node of a synthesised netlist, and whether it reaches the spike output depends on Boolean structure, on what state the neuron is in, and how many cycles pass before it propagates. To see what the architecture can withstand, we must gate-level inject under a realistic workload.

A. Rationale and Toolchain

Two standard approaches were dismissed. Behavioural injection into a source RTL variable operates at the wrong level: the synthesis restructures the arithmetic aggressively, so a 26-bit adder becomes a carry chain with no counterpart in the source, and the gate-level masking that decides survival is invisible above the netlist. Simulator-resident FORCE/RELEASE strikes the right level but does not scale well, and is vendor-specific. Open frameworks [44] pursue the bottleneck with hyperscale compute, but injection remains external to the design; we inject internally.

Since the framework compiles the fault into the circuit itself, there are no interventions at run-time. The framework is implemented with open tooling in five stages (Fig. 7). Each core is elaborated by GHDL and synthesised by Yosys, with the circuit fully flattened so all nets are addressable, then mapped to generic primitives instead of a vendor library, so the reported distribution remains a property of the architecture. In addition, a parser maps each primitive output to one of four semantic zones, namely STATE_MEMORY, CONTROL_PLANE for the nets originated by the threshold comparison or spike signal, DATAPATH for the wide arithmetics, and COMB_CLOUD for everything else. This is not for aesthetic reasons: a datapath defect affects a quantity the loop tracks and can compensate, whereas a control-plane defect destroys a signal the loop depends on.

B. Saboteur Weaving by Logic Shadow Hook

Instrumentation simplifies the engineering burden for a comprehensive campaign: the instrumentor modifies the netlist only once rather than once per fault. At every target the instrumentor inserts a controllable bypass through a three-port fault bus (`fi_enable`, `fi_target_id`, `fi_value`). Each target net is disconnected from its primitive driver, its outputs switched to a shadow net, and the net is re-specified as a continuous assignment selecting between the two:

```
assign net = (fi_enable && fi_target_id==ID)
                  ? fi_value : hook_net;
```

Three properties follow. If `fi_enable` is low then the expression is just the original driver so the instrumented netlist is functionally identical to the golden netlist, and only one design serves the entire campaign. The `fi_target_id` is a top-level input not a constant therefore optimisation cannot remove the multiplexer - a silent failure mode that would make naive netlist mutation fail. And `fi_value` is four state so the hook can force `x` or `z` as easily as a logic level.

C. Simulation and Neuro-Metric Classification

The instrumented netlist is simulated using Icarus Verilog, driven from Python using Cocotb: the latter's interface supports clocks, stimulus and sampling, but it is not the injection mechanism, which is a structural feature. We use a four-valued IEEE 1364 logic and explicitly simulate the unknown state propagation - it's important when a defect feeds a comparator or select line, as a two-valued simulator would silently select a branch and produce a more optimistic result than the hardware delivers. Each run is fault free for 2000 cycles to give the state time to reach its limit cycle, then the fault is asserted permanently, running to cycle $10\,000$ - a timescale made necessary by the 64-cycle leak constant.

A spiking neuron cannot be graded pass or fail: a train displaced by a few cycles still carries its information, whereas

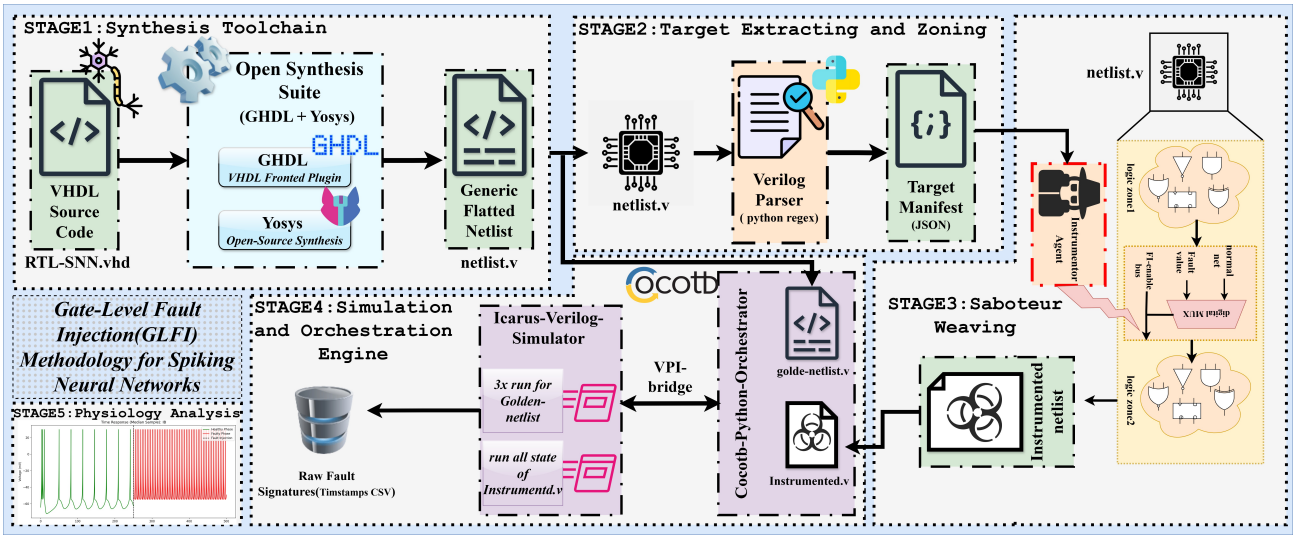

Fig. 7. The five-stage GLFI framework: GHDL/Yosys synthesis to a flattened technology-independent netlist; extraction with semantic zoning; saboteur weaving by Logic Shadow Hook; Cocotb orchestration over Icarus Verilog; and neuro-metric classification of the spike timestamps.

one preserving the count while degenerating into irregular bursts does not. Classification uses the firing-rate error and the ISI coefficient of variation,

$$\mathrm{FRE} = \frac{|N_{\mathrm{faulty}} - N_{\mathrm{golden}}|}{N_{\mathrm{golden}}}, \qquad \mathrm{CV} = \frac{\sigma_{\mathrm{ISI}}}{\mu_{\mathrm{ISI}}} \tag{24}$$

with the tolerance band $\mathrm{FRE} \leq 20\%$ and $\mathrm{CV} \leq 0.20$. An absolute bound on CV is admissible here because the synthesised instance is configured for tonic firing ($c_{hw} = 0$, $d_{hw} = 0.094$), whose golden train is regular and whose fault-free CV sits well inside the band; a bursting configuration such as IB or CH would need the criterion expressed relative to its own golden statistic. Each fault falls into one of four outcomes: *masked*, timestamps identical to the reference; *tolerant*, altered but inside the band; *critical*, outside the band while still firing; and *fatal*, permanently silenced or saturated. The last two are operationally distinct—a critical fault degrades what a neuron contributes, a fatal one removes it or turns it into a source of propagating noise.

D. Fault Model and Scope

The campaign applies the permanent single stuck-at model to all extracted sites, in both polarities, with no fault collapsing. Permanent faults are more difficult for an adaptive scheme, because a stuck-at node influences all later evaluations, rather than one update [37]. Three limits guide the numbers. The data reflect coverage under a given workload, not ATPG fault efficiency [28], and the only observation point is the spike output, so a defect leaving an unchanged emitted train counts as masked. The 10^{-13} residual of Section II-C is the error in the affine map, not of the implemented datapath, which includes the piecewise-linear fit of Section IV-B. And the stability theorem of Section III-A models a defect as a current to the membrane equation, covering DATAPATH and COMB_CLOUD, but not the decision logic that the loop is computed from - the gap the hardened variants of Section IV-D exist to close.

VI. Case Study: On-Chip Learning

The homeostatic machinery which renders H-ASIZH fault-tolerant intervenes exactly on the state variables that synaptic learning would normally alter, so a core hardened in this manner could have sacrificed the plasticity that learning would require. This section settles that question - a confidence check rather than a second contribution.

Fig. 8 shows the system. A $256 \to 1024 \to 10$ network is built entirely from H-ASIZH cores: a hidden layer trained by unsupervised non-causal STDP under first-K temporal k-WTA ($K = 3$), and a ten-neuron output layer trained by a delta rule with teacher annealing. Weights are 4-bit; MNIST digits are reduced to 16×16 and encoded as time-to-first-spike (TTFS). The network is time-multiplexed onto two physical cores, so neuron instances account for 4800 cells while 2.26 Mib of memory dominates the rest, and the mitigation of Section II-D is carried at negligible system cost: the two hardened cores draw $182.8\,\mu$W together, or $0.177\,\mu$W per logical neuron. The organisation cuts both ways, and the trade should be stated: amortising one core over hundreds of virtual neurons amortises the protection, but a single permanent defect in that core then reaches every neuron mapped onto it—which is precisely why the core rather than the array is the right unit to harden.

One point is worth making. Most on-chip learning demonstrations so far have resorted to rate coding or rate-like spike-driven rules [45], and a rate code does not care about small errors in the timing. Temporal coding does care: a TTFS code encodes everything in the order of the spikes, so any change in latency corrupts the code directly - which is why an adaptive-threshold neuron is the right substrate for it. We are not aware of a prior fully end-to-end synthesisable demonstration that combines TTFS coding with on-chip unsupervised learning and a fault-accommodating neuron. The problem is simplified, and is not offered as a benchmark result.

This scope should not be exaggerated. We did not run—and we don't claim to have run—a network-fault campaign: a spiking accelerator is heterogeneous—weight and state mem-

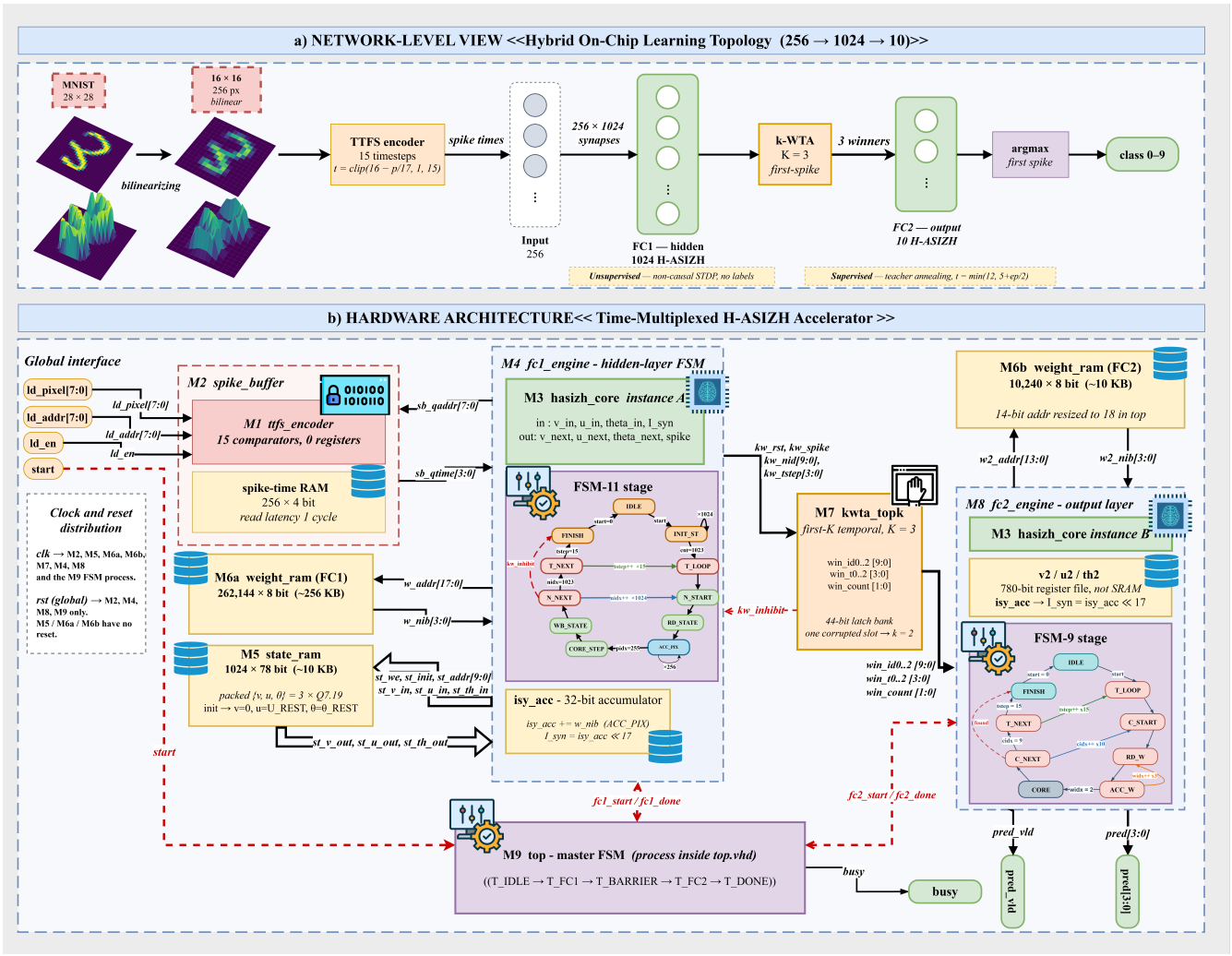

Fig. 8. Case-study system. (a) Network: a $256 \to 1024 \to 10$ feed-forward SNN with TTFS coding over fifteen timesteps, an unsupervised hidden layer under first-K temporal k-WTA and a first-spike output layer. (b) Hardware: the time-multiplexed accelerator, all 1034 logical neurons evaluated on two physical H-ASIZH cores.

ories, address generator, winner arbitration, control FSMs—every part must be mitigated, so claiming system reliability at the neuron level is unsound. The campaign of Section V is exclusive to H-ASIZH, the only unit for which our numbers hold.

VII. Experimental Results

Five descriptions were passed through the same flow and Campaign of Section V: the unprotected *Baseline*, the intrinsic *Homeostatic* core, that core with *Light TMR* on the control plane, the same core with the *Restoring Organ*, and a fully triplicated reference.

A. Physical Implementation and PPA

All five VHDL cores were elaborated by GHDL, synthesised by Yosys, placed, clock-tree synthesised and routed using OpenLane on SkyWater's open 130 nm PDK. OpenSTA signed off timing and power at the typical corner (1.8 V, 25 °C) on the netlist after routing with parasitic extraction, meaning delays capture loading of interconnect.

A single $80\,146.87\,\mu\text{m}^2$ floorplan was fixed throughout, so the die outline is not the discriminating metric; occupied silicon is, and it follows utilisation, rising from 22.35% to 75.98%—from $17\,913$ to $60\,896\,\mu\text{m}^2$ of standard cells (Table II). The intrinsic controller is inexpensive: threshold state, leak, penalty accumulator and brake cost 697 cells ($1.39\times$) and $1.72\times$ the power. Triplication compounds, multiplying cells by $3.40\times$ and power by $3.80\times$—the latter larger because replicated logic also multiplies switching activity—with wire length up $4.12\times$. The critical path lengthens from 11.96 to 15.04 ns, the threshold comparison now resolving against a register updated in the same cycle, placing $F_{\max}$ at 64.92 MHz. Full triplication is the faster circuit at 80.00 MHz, since it replicates the baseline datapath and adds only a voter; the next paragraph shows that advantage does not survive translation into energy.

Absolute dissipation stays inside the budget the application demands: $30.7\,\mu$W unprotected and $91.4\,\mu$W hardened at the typical corner, so a defect-tolerant Izhikevich neuron costs under a tenth of a milliwatt. Leakage is negligible throughout—between 6.9 and 7.3 pW per cell, never above 0.05% of the

TABLE II
PHYSICAL IMPLEMENTATION SUMMARY (SKYWATER 130 NM, OPENLANE P&R, OPENSTA TYPICAL-CORNER SIGNOFF)

Metric	1. Baseline	2. Homeostasis	3. + Light TMR	4. + Restoring Organ	5. Full TMR
Core area (μm^2, fixed floorplan)	80 146.87	80 146.87	80 146.87	80 146.87	80 146.87
Placement utilisation	22.35%	30.26%	50.03%	51.61%	75.98%
Standard-cell area (μm^2)	17 913	24 252	40 098	41 364	60 896
Standard-cell count	1 772	2 469	3 969	4 093	6 025
Cell overhead	1.00×	1.39×	2.24×	2.31×	3.40×
Critical-path delay (ns)	11.96	15.04	15.58	16.10	12.50
$F_{\max}$ (MHz) = throughput (M updates/s)[a]	83.61	64.92	64.18	62.11	80.00
Internal / switching power (μW)	13.4 / 17.3	22.6 / 30.2	36.3 / 52.2	37.4 / 54.0	45.6 / 71.2
Leakage power (nW)	13.00	17.00	27.32	28.17	44.20
Total core power (μW)[b]	**30.7**	**52.8**	**88.5**	**91.4**	**116.8**
Power overhead	1.00×	1.72×	2.88×	2.98×	3.80×
Energy per neuron update (pJ)	**0.367**	**0.813**	**1.379**	**1.472**	**1.460**
Total wire length (μm)	59 207	87 411	151 221	156 465	243 932
Interconnect vias	14 185	19 827	31 861	32 853	48 229

[a] $F_{\max}$ is the sign-off frequency reported by OpenSTA with clock uncertainty and skew included, and is therefore slightly below the reciprocal of the raw critical-path delay.
[b] Total = internal + switching. Leakage stays below 0.05% of the total in every configuration (6.9–7.3 pW per standard cell), so it is reported separately rather than added.

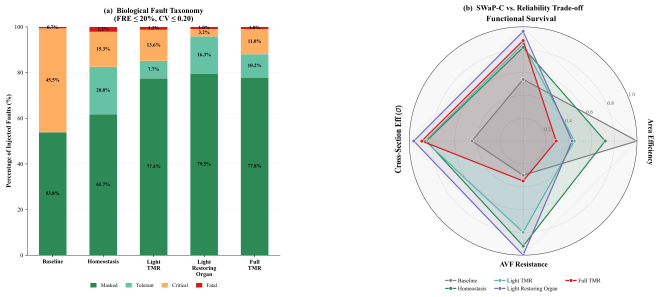

Fig. 9. Campaign results for the five cores. (a) Gate-level fault outcomes under the criteria of Section V-C, tolerance band FRE $\leq$ 20%, CV $\leq$ 0.20; the tolerant category appears only once adaptive thresholding is present. (b) Joint reliability and SWaP-C trade-off, each axis normalised to its best observed value.

total—which is what a 130 nm high-density library gives at 25 °C and confirms the figures are dynamic-dominated.

Frequency and power compose into the metric that matters for an event-driven fabric, the energy of one update, $E = P_{\text{total}}/F_{\max}$. The unprotected core spends 0.367 pJ, the homeostatic core 0.813 pJ. The comparison that settles this paper's argument sits at the other end of the table: the restoring-organ core spends 1.472 pJ against 1.460 pJ for full triplication—the same number to within 1%. Triplication is faster and no more efficient, yet leaves 12.0% of defects dangerous where the proposed core leaves 4.2% (Section VII-B). At equal energy per update the accommodating core is roughly three times safer.

B. Reliability and the Cost Trade-off

Fig. 9(a) shows the results of the distribution of outcomes, with every node injected in both polarities. The unprotected core defines the magnitude of the problem: 45.5% of faults go out of the admissible envelope, while 0.7% are fatal, and zero outcomes are tolerant (without a self adaptive threshold, a fault is either totally invisible or destructive, and graceful degradation is not something the architecture has).

The controller changes the shape of that distribution rather than shifting it. Critical outcomes fall threefold to 15.3% and a tolerant population of 20.8% appears where none existed, the accommodation the Lyapunov analysis predicts. The same data expose its limit: fatal outcomes rise from 0.7% to 2.2%, because sustained discharge accumulates inhibitory charge through the brake, which relaxes only with the intrinsic time constant—a controller able to suppress a seizure can, by the same action, suffocate the cell. The restoring organ addresses exactly that: critical falls to 3.2%, fatal returns to 1.0%, and masked reaches 79.5%, for 124 cells over light TMR.

Fig. 9(b) places the five on the joint reliability and SWaP-C surface. Taking the dangerous fraction as critical plus fatal, the restoring-organ core leaves 4.2% against 12.0% for full triplication, with 32% fewer cells, 22% less power and 36% less wire. It's structural: spatial redundancy outvotes a fault, does nothing about a defect all three copies see equally, expands the circuit, but accommodation recomposes a steady state in the presence of the defect, regardless of how many copies observe it. The same reading separates this work from its bio-inspired forebears. The SPANNER [16] and astrocytic cluster [17], [18] monitors are placed next to the neuron, built from the same gates on the same die, such that a defect in the monitor itself removes or reverses the protection. Closest to this work, Johnson *et al.* [19] achieve homeostasis from another rule-holder, a distributed set of astrocytic cells, so regulator and process do not share the same circuit. Here, the regulated variable, sensor and actuator are the same three registers the neuron already needs, so no supervisory fabric is left to fail. The dendritic family [21] acts inside the neuron as well, but is resilient structurally at overhead almost 60%, against 39% here without replicas; migration schemes [24], [25], [39] skip overhead but render a spike's path a trait of the die. The complementarity runs deeper, where upset-

TABLE III
RELIABILITY FIGURE OF MERIT (RFOM) FROM THE CAMPAIGN. THE FINAL COLUMN IS THE QUOTIENT OF THE TWO PRECEDING ONES

Architecture	Cells	AVF	Vuln. reduction	Area growth	RFoM
Baseline	1 772	0.462	1.00×	1.00×	1.00×
Homeostasis	2 469	0.175	2.64×	1.39×	1.89×
+ Light TMR	3 969	0.149	3.10×	2.24×	1.38×
+ Restoring Organ	4 093	0.042	11.00×	2.31×	**4.76**×
Full TMR	6 025	0.120	3.85×	3.40×	1.13×

tolerant cells [33] resist changes to the state a neuron maintains rather than the computations it performs, and further away, where approximate memories trade inference energy against a contained error budget [36].

One number is the same for two very different architectures: 1.0% fatal fraction for restoring-organ core and for full triplication. The faults are on the spike output and the nets driving it, after the voter. And a voter corrects an error amongst its inputs but cannot correct its own output. But the floor is that of the observation boundary and not that of the architecture; reaching it at 4 093 cells where triplication requires 6 025 is the pragmatic statement of efficiency.

C. Throughput and Time Multiplexing

The signed-off 64.92 MHz is lower than figures quoted for comparable cores, but the relevant measure is sustained throughput $\Theta = f_{\mathrm{clk}}/\mathrm{CPU}$, with CPU the initiation interval between updates: a core at 200 MHz with $\mathrm{CPU} = 4$ delivers 50 M updates/s, less than this design, where $\mathrm{CPU} = 1$ by construction. Nor is the frequency an architectural bound—splitting the 15.04 ns path into S stages at $T_{\mathrm{ov}} \approx 0.4$ ns each gives roughly 235 MHz at $S = 4$ and 430 MHz at $S = 8$. Pipelining ordinarily raises CPU in proportion and breaks the recurrence; time multiplexing resolves both, and is the organisation this core targets—one datapath cycling through many virtual neurons, each still seeing a single-cycle recurrence in its own timebase. What matters for an array is then

$$N_{\mathrm{virtual}} = \frac{\Theta}{f_{\mathrm{bio}}} = \frac{f_{\mathrm{clk}}}{\mathrm{CPU} \cdot f_{\mathrm{bio}}} \tag{25}$$

At the conventional 1 ms timestep the signed-off clock sustains about 64 900 neurons per core, meeting real time with four orders of magnitude of margin.

D. Reliability Figure of Merit

The distributions above give the architectural vulnerability factor, $\mathrm{AVF} = (N_{\mathrm{critical}}+N_{\mathrm{fatal}})/N_{\mathrm{total}}$, which is not by itself a comparison between architectures: a mitigation that lowers it also enlarges the circuit in which defects can occur. Under a single permanent defect, independent and uniform over cells, the expected number of dangerous events scales as the product and the per-cell rate cancels in a ratio, so we score a mitigation by

$$\mathrm{RFoM}_i = \underbrace{\frac{\mathrm{AVF}_{\mathrm{base}}}{\mathrm{AVF}_i}}_{\text{vulnerability reduction}} \Big/ \underbrace{\frac{N_i}{N_{\mathrm{base}}}}_{\text{area growth}} \tag{26}$$

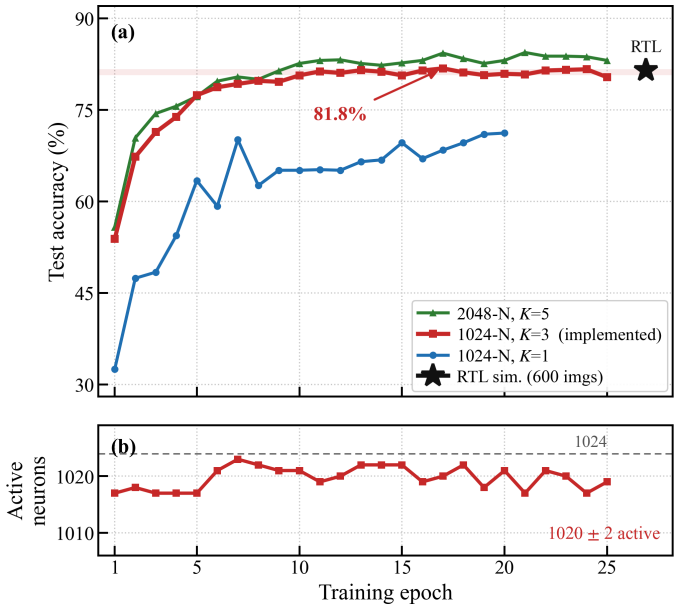

Fig. 10. On-chip learning on the network of Fig. 8. (a) Test accuracy against epoch for three design points; the band is the steady-state mean $\pm\sigma$ over the final five epochs of the implemented $K = 3$ core, whose peak epoch reaches 81.8% and whose steady state is $81.2 \pm 0.5\%$. The star marks the RTL simulation over 600 images. (b) Active hidden neurons per epoch.

the reliability figure of merit. A mitigation is worth its silicon only when $\mathrm{RFoM} > 1$. It is a first-order screening ratio rather than a lifetime, since absolute mean times to failure need foundry defect densities the open PDK does not supply, and it charges triplication under the single-defect model the campaign implements; a full lifetime treatment of a voted structure is second order in defect probability and follows the classical $R_{\mathrm{TMR}}(t) = 3R^2(t) - 2R^3(t)$ analysis, which retains an advantage this metric does not credit at very low defect density.

Table III evaluates the five. Full triplication cuts propagation probability by 3.85× yet returns only $\mathrm{RFoM} = 1.13$, because that reduction required 3.40× the cells and the enlarged target consumes almost the whole benefit. The restoring-organ core gives $\mathrm{RFoM} = 4.76$ from 11.00× vulnerability reduction against 2.31× area growth, and even the plain homeostatic core outperforms full triplication by 1.67× at 41% of its cell count. Replication is self-limiting by construction: every copy that lowers the vulnerability factor raises the count that multiplies it, whereas homeostasis lowers AVF without replicating anything.

E. On-Chip Learning

The network of Section VI was trained for 25 epochs on 10 000 images and evaluated on 2000. Fig. 10(a) gives the outcome: the implemented configuration reaches 81.8% and holds $81.2 \pm 0.5\%$ over its final five epochs, while strict $K = 1$ competition converges more slowly to a markedly lower plateau. The synthesisable RTL, simulated end to end over 600 test images, returns 81.5%—within one standard error of the software model, so the fixed-point datapath is a faithful realisation of it.

Fig. 10(b) reports the most relevant outcome: Across all 25 epochs, 1020 ± 2 of the 1024 hidden neurons remain active. Both neuron death and domination are classic unsupervised competitive-learning failure modes; the adaptive threshold suppresses both - the same mechanism that stops a faulted neuron from runaway also stops an untrained one from being silenced. The Section II-D hardening does not just preserve learning; it is what makes the population viable until learning is complete.

VIII. Conclusion

This paper set out to obtain fault tolerance for a digital Izhikevich neuron at minimal overhead, and did so by moving the mechanism inside the neuron: the firing threshold becomes a slow state variable whose negative feedback accommodates a permanent defect, proven Uniformly Ultimately Bounded and realised in shifts and additions on a single-cycle Q7.19 datapath at 30.7–$91.4\,\mu$W. An exhaustive gate-level campaign leaves 4.2% of faults dangerous against 12.0% for full triplication, at 32% fewer cells, a $4.76\times$ reliability figure of merit against $1.13\times$, and—the comparison that carries the argument—the same 1.47 pJ per neuron update. Triplication buys nothing that accommodation does not buy more cheaply, because every replica that lowers the vulnerability factor enlarges the target that multiplies it.

In another way, this mechanism is cheaper than we'd thought. The threshold-adapting neuron can't runaway or be permanently outcompeted, so the three registers that absorb the defect also suppress neural death and dominance in unsupervised learning: 1020 of the 1024 hidden neurons remain active through training. This isn't a fault-tolerance result; it's due to the learning dynamics, and we report it as such, but the silicon bought for high-reliability is active even without a fault.

The way forward is composition rather than extension. Intrinsic accommodation and system-level protection are complementary: the former eliminates the neuron as a single point of failure, ECC-protected memories, control-path duplication and checkpoint-and-retry address the parts of an accelerator that a neuron model cannot reach. Extending the treatment to the memory hierarchy, the arbitration logic and the control FSMs are topics of current research.

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